

Assignment 1

Choose 7 of the 11 problems below that you would happy to present at the **Fri Oct 16th Problem session**.

Deadline: email which 7 questions these are to fiona.skerman@math.uu.se and anda.skeja@math.uu.se before the lecture on **Tue Oct 13th**. We will let you know on Tue which question you will present on Fri.

The following inequalities may be useful.

- $1 + x \leq e^x$ for all $x \in \mathbb{R}$.
- $(1 + a)^n \leq e^{an}$ for $a > -1$, $n \geq 0$.
- More terms: $1 - x = e^{-x - \frac{x^2}{2} - \frac{x^3}{3} - \dots - \frac{x^k}{k} + O(x^{k+1})}$, and $1 + x = e^{x - \frac{x^2}{2} + \frac{x^3}{3} \pm \dots + O(x^{k+1})}$ as $x \rightarrow 0$
- Stirling formula: $n! \sim \sqrt{2\pi n} \frac{n^n}{e^n}$. Simpler formula that is often enough: $k^k/e^k \leq k! \leq k^k$
- $\left(\frac{n}{k}\right)^k \leq \binom{n}{k} \leq \frac{n^k}{k!} \leq \left(\frac{en}{k}\right)^k$ for $1 \leq k \leq n$.
- If $k = k(n) = o(\sqrt{n})$, then $\binom{n}{k} \sim \frac{n^k}{k!}$ as $n \rightarrow \infty$

1. Let Σ be a finite set of 0-1-strings with lengths

$$\ell_1, \ell_2, \dots, \ell_k.$$

Suppose that Σ is *prefix-free*, meaning that no string in Σ is a prefix of another. Prove that

$$\sum_{i=1}^k 2^{-\ell_i} \leq 1.$$

2. Assume that for all integers $n \geq k \geq 1$ we have $R(k, k) > n - \binom{n}{k} 2^{1 - \binom{k}{2}}$. Show that one may set $n = \lfloor e^{-1} k 2^{k/2} \rfloor$ to obtain

$$R(k, k) \geq \frac{k 2^{k/2}}{e} (1 - o(1))$$

3. A graph H is r -partite, if its vertex set can be partitioned into r parts $\{V_1, \dots, V_r\}$ (where we allow $V_i = \emptyset$), such that each part contains no edges, i.e. $e(V_i) = 0$ for $i = 1, \dots, r$. (Equivalently, the chromatic number satisfies $\chi(H) \leq r$).

Fix a graph G . For any $r \geq 2$, show that G contains an r -partite subgraph H with $e(H) \geq e(G)(r-1)/r$.

4. Let $G = (V, E)$ be a graph with n vertices and m edges, and denote by d_v the degree of a vertex v .

- (a) Consider a random ordering of the vertex set V (chosen uniformly at random from all $n!$ possible orderings). What is the probability that a given vertex v comes before all of its neighbours?
- (b) Show that G has an independent set (a set of vertices without any edges between them) of size at least $\sum_{v \in V} \frac{1}{d_v + 1}$.
- (c) Deduce that G has an independent set of size at least $\frac{n^2}{2m+n}$.
5. Show that $\frac{\log n}{n}$ is a threshold function in $G(n, p)$ for the property that every vertex in the graph has degree at least 1.
- Hint:* Apply the first and second moment method to the number of isolated vertices. A useful approximation: For n and $k(n)$ so that $|k| = o(\sqrt{n})$ as $n \rightarrow \infty$, we have $(1 + \frac{k}{n})^n \sim e^k$ as $n \rightarrow \infty$.
6. First a definition. A sum $S = \sum_{i \geq 0} a_i$ which is either finite or infinite and convergent is said to satisfy the alternating inequalities if the partial sum $\sum_{i=0}^t a_i$ is at least S for all even t and at most S for all odd t ; that is, the partial sums alternately *over-* and *under-estimate* the final result.

Now, suppose $I_1, \dots, I_n$ are the indicator functions of n events $E_1, \dots, E_n$. For $0 \leq r \leq n$ let $S_r = \sum_{A \subseteq [n], |A|=r} \prod_{i \in A} I_i$, where $[n] = 1, 2, \dots, n$.

- (a) Show that

$$\prod_{i=1}^n (1 - I_i) = \sum_{r=0}^n (-1)^r S_r,$$

and that the sum satisfies the alternating inequalities. (*Note:* Both sides are random; the statement is that the relevant inequalities always hold. You may want to consider different cases according to how many of the events E_i hold.)

- (b) Deduce that

$$P(\text{no } E_i \text{ holds}) = \sum_{r=0}^n (-1)^r \sum_{A \subseteq [n], |A|=r} P(\cap_{i \in A} E_i)$$

and that the sum satisfies the alternating inequalities. (This is a form of the inclusion-exclusion formula.)

7. Let X be a random variable taking values in $\{0, 1, \dots, n\}$, define $\mathbb{E}_r[X] = \mathbb{E}[X(X-1)\dots(X-r+1)]$ (the r -th falling factorial moment). Show that

$$P(X=0) = \sum_{r=0}^n (-1)^r \frac{\mathbb{E}_r[X]}{r!}$$

Lovász Local Lemma:

8. Show that it is possible to colour the edges of K_n with $k = \lceil 3\sqrt{n} \rceil$ colours so that no triangle is monochromatic.
9. A k -SAT formula is an expression such as (for $k = 3$)

$$(x_1 \text{ OR } x_4 \text{ OR } \overline{x_6}) \text{ AND } (\overline{x_2} \text{ OR } x_3 \text{ OR } \overline{x_4})$$

$$\text{AND } (\overline{x_1} \text{ OR } \overline{x_2} \text{ OR } \overline{x_5}) \text{ AND } (x_2 \text{ OR } x_3 \text{ OR } x_6)$$

where the variables x_i take values TRUE or FALSE, $\overline{x_i}$ means NOT x_i , and in each clause k distinct variables x_i or their negations are OR'd together. A formula is *satisfiable* if there is an assignment of TRUE/FALSE-values to the variables x_i which makes the expression true.

- (a) Use the first moment method to show that any k -SAT-formula with fewer than 2^k clauses is satisfiable.
 - (b) For $k \geq 4$, use the Symmetric Local Lemma to show that any k -SAT-formula in which no variable appears in more than $2^{k-2}/k$ clauses is satisfiable.
10. Let $G = (V, E)$ be a graph with maximum degree Δ , and let $V_1, \dots, V_s$ be a collection of disjoint subsets of V , each of size $k \geq 2e\Delta$. Show that G has an independent set which contains a vertex from each set V_i .

Hint: Pick one vertex X_i from each set V_i . You should consider a ‘bad’ event E_{uv} for each edge uv with u and v in different sets V_i, V_j .

Harris’s Lemma:

11. Let $X = \{x_1, \dots, x_n\}$ be a finite set, $p_1, \dots, p_n \in [0, 1]$, and let $\mathbb{P}$ be the probability measure associated with choosing a random subset of X by including every element x_i independently with probability p_i .

Let $\mathcal{U}_1, \dots, \mathcal{U}_k \subset \mathcal{P}(X)$ be up-sets (increasing events), and $\mathcal{D}$ a down-set.

- (a) Show that $\mathbb{P}\left(\bigcap_{i=1}^k \mathcal{U}_i\right) \geq \prod_{i=1}^k \mathbb{P}(\mathcal{U}_i)$.
- (b) Show that $\mathbb{P}(\mathcal{U}_1 \mid \mathcal{D}) \leq \mathbb{P}(\mathcal{U}_1)$.
- (c) Show that if $\mathcal{U}_1$ and $\mathcal{U}_2$ are independent, then $\mathbb{P}(\mathcal{U}_1 \mid \mathcal{U}_2 \cap \mathcal{D}) \leq \mathbb{P}(\mathcal{U}_1 \mid \mathcal{U}_2)$.
- (d) Does the last inequality always hold, i.e. without assuming independence?